Seat No._____

GANPAT UNIVERSITY

B. Sc. IT (DATA SCIENCE) SEM-I EXAMINATION (CBCS)

U71A1IP1: INTRODUCTION TO PROGRAMMING-I

Nov-Dec 2022

[Time: 3 Hours] **[Total Marks: 60]**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any Five) **(30)**

a) What is Flowchart? Discuss various Flowchart Symbols.

b) Explain basic structure of C Program with example.

c) Write a short note on Symbolic Constants.

d) What is Logical Operator? Write a C program to find smallest number among three numbers.

e) Discuss switch case statement with proper example.

f) Explain while loop with example.

g) Write a C Program to display Fibonacci series between 1 to 100 numbers.

SECTION-II

2. Answer the following. (Any Two) **(12)**

a) List out Types of Algorithm. Discuss Looping Type of Algorithm with example.

b) Write an Algorithm and draw a Flowchart to display reverse number of a given number.

c) Write a short note on Primary Data Types.

3. Answer the following. (Any Two) **(12)**

a) Discuss Increment and Decrement operators with example.

b) Explain Nested if… else statement with proper example.

c) Write a short note on goto statement with example.

4. Answer the following. (Any One) **(06)**

a) What is looping? Explain for loop with proper example.

b) Discuss break and continue statement with example.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY

B.Sc (IT) (DS) SEM-I EXAMINATION

Nov-Dec 2022

U71A2DA1: DATA ANALYSIS-I

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. **Answer the following: (Any Five)** **(30)**

1. What is documentation? Explain different views of document.
2. Describe the ways of Addressing Modes with example.
3. Write a short note on Conditional Formatting of Spreadsheet.
4. How to find and replace text in document? Write down the step for using find and replace in document.
5. Describe the MS-Access Datatype.
6. Write a short note on Slide Transition.
7. Discuss Custom Animation in detail.

SECTION-II

2. **Answer the following: (Any Two)** **(12)**

1. What is Mail Merge? Define step of Mail Merge.
2. Discuss the Naming Range with suitable example.
3. What is chart? Explain any two charts in detail.

3. **Answer the following: (Any Two)** **(12)**

1. Explain different views of Presentation.
2. Define the following term:
   a) Freeze Pane
   b) Auto Filter
   c) Drawing Toolbar
3. What is Data Validation? Explain the use of Data Validation.

4. **Answer the following: (Any One)** **(06)**

1. What is MS Access Database? Define the steps for creating Database.
2. Describe the Different elements of database.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY

B.Sc. IT (DATA SCIENCE) SEM-I EXAMINATION

Nov-Dec 2022

U71A3WD1: WEB DESIGNING-I

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any Five) **(30)**

[A] What is URL? Explain the structure of URL in brief.
[B] State the importance of Web designing principles.
[C] Explain the HTML web page structure in details.
[D] Discuss any five New Semantic Tags in HTML 5.
[E] Explain the Element of List Tags in details with example.
[F] Explain CSS background properties with proper example.
[G] Explain the JavaScript operators in details.

SECTION-II

2. Answer the following. (Any Two) **(12)**

[A] State the difference between Static & Dynamic Web sites.
[B] State the difference between Internets, Intranet & Extranet.
[C] Discuss the Table tag with its all elements and example.

3. Answer the following. (Any Two) **(12)**

[A] Explain CSS Selectors with example.
[B] Writes HTML code for the creation of Student registration form.
[C] State the importance of CSS with its advantage.

4. Answer the following. (Any One) **(06)**

[A] How can we perform looping in JavaScript? Explain with example.
[B] Elaborate the JavaScript function with example.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc.IT (Data Science) SEM-I EXAMINATION
Nov-Dec 2022
U71A5FM FUNDAMENTAL OF MATHEMATICS

[Time: 3 Hours] **[Total Marks: 60]**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any Five) **(30)**

1. Let $A = \{1,2\}, B = \{4,5\}, C = \{5,6\}$ then evaluate

1) $A\,X\,(B \cup C)$,
2) $A\,X\,(B \cap C)$,
3) $(A\ X\,B) \cup (B\,X\,C)$.

2. Define the following set by Roaster method

1) $A = \{x/x \text{ is a single digited prime number}\}$
2) $B = \left\{x/x = \frac{n}{n+1}, n \in N, n \leq 4\right\}$
3) $C = \{x/\ x^2 - 7 = 0, x \epsilon R\}$

3. Define following terms:

Matrix, Null Matrix, Row Matrix, Symmetric Matrix, Lower Triangular.

4. If $A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 4 & -4 \\ 5 & 2 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 & 2 \\ -8 & 4 & 6 \\ -5 & 3 & 2 \end{bmatrix}$ then find $A + B$ and A – B.

5. Discuss Isomorphic Graph in detail.

6. Define the following terms in context of graph theory with suitable example: Simple graph, Multi graph, loop, Open Walk, Isolated Vertex.

7. Find the number of words, with or without meaning that can be formed with the letters of the word 'SWIMMING?

SECTION-II

2. Answer the following: (Any Two) **(12)**

1. Three newspaper A, B, C are published from a city. The survey of 200 people in the city provided the following information :

1) 88 People read A.
2) 82 People read B.

3) 68 People read C.

4) 28 People read both A and B.

5) 46 People read both B and C.

6) 23 People read both C and A.

7) 8 People read all the three papers.

Answer the following:

1) Out of 200 people, how many are reading all least one paper?

2) How many people read none of these papers?

3) What is the number of people who read exactly two papers?

2. If $A = \begin{bmatrix} 2 & 7 & 4 \\ 0 & 3 & -4 \\ 5 & 2 & 12 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 2 \\ -8 & 4 & 4 \\ -5 & 3 & 1 \end{bmatrix}$ then find BA and AB.

3. If $A = \begin{bmatrix} 2 & -5 \\ 3 & 1 \end{bmatrix}$ then find A^2 and A^3.

3. **Answer the following: (Any Two)** **(12)**

1. Determine whether the following functions are one-one or many one
(i)$f: N \to N, f(x) = x + 5$ (ii) $f: R \to R, f(x) = |x| + 1$

2. If $f(x) = \begin{cases} 2 + 3x \ , \ x \geq 0 \\ 2 - 3x \ , \ x < 0 \end{cases}$

Then find $f(2) + f(0) + f(-2) + f(3) - f(5)$

3. Discuss homographic Graph in detail.

4. **Answer the following: (Any One)** **(06)**

1. Find the number of different words that can be formed with the letters of the word 'BUTTER' so that the vowels are always together.

2. How many 4 digit numbers that are divisible by 10 can be formed from the numbers 3, 5, 7, 8, 9, 0 such that no number repeats?

END OF PAPER

Enroll No._____

GANPAT UNIVERSITY

B.Sc. (DS) SEMESTER-I EXAMINATION (CBCS)

Nov-Dec 2022

U71B6CS1: LANGUAGE & COMMUNICATION SKILLS-I

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

Q:1. **Attempt any five out of seven:** **(30)**

1 Discuss how physical and psychological barrier affect the communication.

2 Explain the difference between hearing and listening.

3 Describe communication with various definitions.

4 7C's plays a vital role in communication, justify this statement in your own words.

5 Write the difference between skimming and scanning with example.

6 Summarize the non-verbal communication in details with its sub parts.

7 List the barriers of listening with an example in your own words.

SECTION-II

Q:2. A. **Fill in the blanks using an appropriate answer from the options given in the brackets:** **(6)**

1 What does refer to the body movements, which support and emphasize the state of mind of an individual.

(A) Gesture (B) Posture (C) Eye contact

2 The worker conveys message to the production manager or director of the company: what type of communication is this?

(A) Horizontal (B) Downward (C) Upward

3 The old woman walked carefully with her heavy load _____ she should fall.

(A) until (B) although (C) lest

4 I gave _____ a cough to catch her attention.

(A) gave (B) took (C) paid

5 There was no _____ in the train.

(A) let up (B) let in (C) let down

6 **Dancing** is her hobby. The underline word use as_____.

(A) Verb (B)gerund (C) participle

Seat No._____

GANPAT UNIVERSITY

B.Sc. IT(DS) SEM-I Regular Theory Nov Feb-2022

January - February 2022

U71B6CS1: LANGUAGE & COMMUNICATION SKILLS-1

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

Q 1. Answer the following: (Any Five) **(30)**

1. Illustrate through a neat diagram the elements of the communication process.
2. What is the Phrasal Verb? Explain the term with appropriate examples.
3. Explain the need and importance of listening in communication. State the barriers to listening and ways to overcome the barriers.
4. Explain in detail the techniques for effective reading comprehension.
5. Oral communication skills form an important part of your personality. Explain the statement with suitable examples.
6. List out and discuss all **4 Cs** of effective written communication.
7. Describe any four Non-verbal modes by giving examples.

SECTION-II

Q.2. A. What is Barrier? Discuss organizational barriers with apt examples. **(06)**

OR

A. Differentiate between formal and informal communication. **(06)**

Q.2. B. Fill in the blanks with appropriate alternatives. **(06)**

1. I want _____ apple from the basket. (a, an, the)
2. He _____ a novel since October, and now he is about to finish it.

   (was writing, has been writing, is writing)
3. This is the place _____ we stayed last time we visited. (where, when, how)
4. If you park there, you will have to _____ a fine.(pay, take, have)
5. The name of a person, place or thing is called_____.(noun, pronoun, adjective)
6. The government has _____all projects.(cast by, cast aside, cast out)

OR

B. Fill in the blanks with appropriate alternatives. **(06)**

1. Swami Vivekananda _____ a disciple of Swami Ramakrishna Pramhans.

   (is, was, were)
2. Shall and will are often used in the _____tenses. (past, present, future)
3. We _____ the longest coastline in Gujarat. (have, has, had)
4. We are going to see _____ Statue of Unity this Diwali. (a, an, the)
5. Komal has _____ unique idea to solve this problem. (a, an, the)
6. A collocation is two or more words that _____.

   (often go together, have similar meanings, must be used together)

Q. 3. A. Answer the following: **(06)**

What is listening? Explain the process of listening.

OR

Highlights all the important aspects of reading, including skimming and scanning. **(06)**

B. Write a short note: (Any One) **(06)**

a) Traits of a good listener

b) Importance of tone, stress and voice modulation in speaking

c) Importance of punctuation marks

Q. 4. Answer the following: (Any One) **(06)**

1. Give the diagrammatic presentation of the semi-block layout form of a letter.
2. Explain the essential elements of a formal letter.

END OF PAPER